

SET THEORY – DETAILED STUDY NOTES

Mathematics / Computer Science Fundamentals

Set Theory is the branch of mathematics that deals with the study of sets — collections of well-defined and distinct objects. It forms the foundation for many topics in mathematics and computer science, including logic, probability, and database theory.

1 ■■ Definition of a Set

A **set** is a collection of distinct objects, considered as a single entity. The objects in a set are called **elements** or **members** of the set. Sets are usually denoted by capital letters (A, B, C, ...) and their elements are listed within curly brackets { }.

Example:

$A = \{2, 4, 6, 8\} \rightarrow$ A set of even numbers less than 10.

$B = \{a, e, i, o, u\} \rightarrow$ A set of vowels in the English alphabet.

2 ■■ Types of Sets

- **Empty (Null) Set:** A set with no elements. Example: $\emptyset$ or $\{\}$.
- **Finite Set:** A set with a countable number of elements. Example: $\{1, 2, 3, 4\}$.
- **Infinite Set:** A set with uncountable elements. Example: $\{1, 2, 3, \dots\}$.
- **Equal Sets:** Two sets with exactly the same elements. Example: $\{1, 2, 3\} = \{3, 1, 2\}$.
- **Equivalent Sets:** Two sets with the same number of elements but not necessarily identical elements.
- **Subset:** A is a subset of B if every element of A is also an element of B. ($A \subseteq B$)
- **Proper Subset:** $A \subset B$ if $A \subseteq B$ and $A \neq B$.
- **Universal Set:** A set that contains all elements under consideration, denoted by U.
- **Power Set:** The set of all subsets of a given set A, denoted by $P(A)$.
- **Complement of a Set:** All elements in the universal set U that are not in A, written as A' or A^c .

3 ■■ Set Notations and Representations

Sets can be represented in two major ways:

- **Roster or Tabular Form:** All elements of the set are listed, separated by commas. Example: $A = \{2, 4, 6, 8\}$.
- **Set-builder Form:** Describes properties of elements. Example: $A = \{x \mid x \text{ is an even number less than } 10\}$.

4■■■ Set Operations

Operations on sets help to combine, compare, and manipulate sets.

- **Union ($A \cup B$):** Set of all elements that are in A or in B or in both.
- **Intersection ($A \cap B$):** Set of all elements that are common to both A and B.
- **Difference ($A - B$):** Set of elements in A but not in B.
- **Complement (A'):** All elements in the universal set U that are not in A.
- **Symmetric Difference ($A \Delta B$):** Elements that belong to A or B but not both.

5■■■ Laws of Set Operations

- Commutative Law: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
- Associative Law: $(A \cup B) \cup C = A \cup (B \cup C)$.
- Distributive Law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
- Identity Law: $A \cup \emptyset = A$ and $A \cap U = A$.
- Complement Law: $A \cup A' = U$ and $A \cap A' = \emptyset$.
- Idempotent Law: $A \cup A = A$ and $A \cap A = A$.
- Domination Law: $A \cup U = U$ and $A \cap \emptyset = \emptyset$.
- De Morgan's Laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

6■■■ Venn Diagrams

Venn Diagrams are visual tools used to represent relationships between sets using circles. Each circle represents a set, and overlaps indicate intersections. They are useful for illustrating unions, intersections, and complements.

7■■■ Solved Example

If $U = \{1,2,3,4,5,6,7,8\}$, $A = \{1,2,3,4\}$, $B = \{3,4,5,6\}$, find:

- (a) $A \cup B$
- (b) $A \cap B$
- (c) A'

Solution:

- (a) $A \cup B = \{1,2,3,4,5,6\}$
- (b) $A \cap B = \{3,4\}$
- (c) $A' = \{5,6,7,8\}$

8■■■ Applications of Set Theory

Set theory is applied in computer science (data structures, databases), logic, probability, and programming. It is foundational in defining relations, functions, and algorithms.